Bocconi

ACCIDENT IN A CO-GENERATION PLANT

Analysis of the Human and Organisational Factors (HOF) involved

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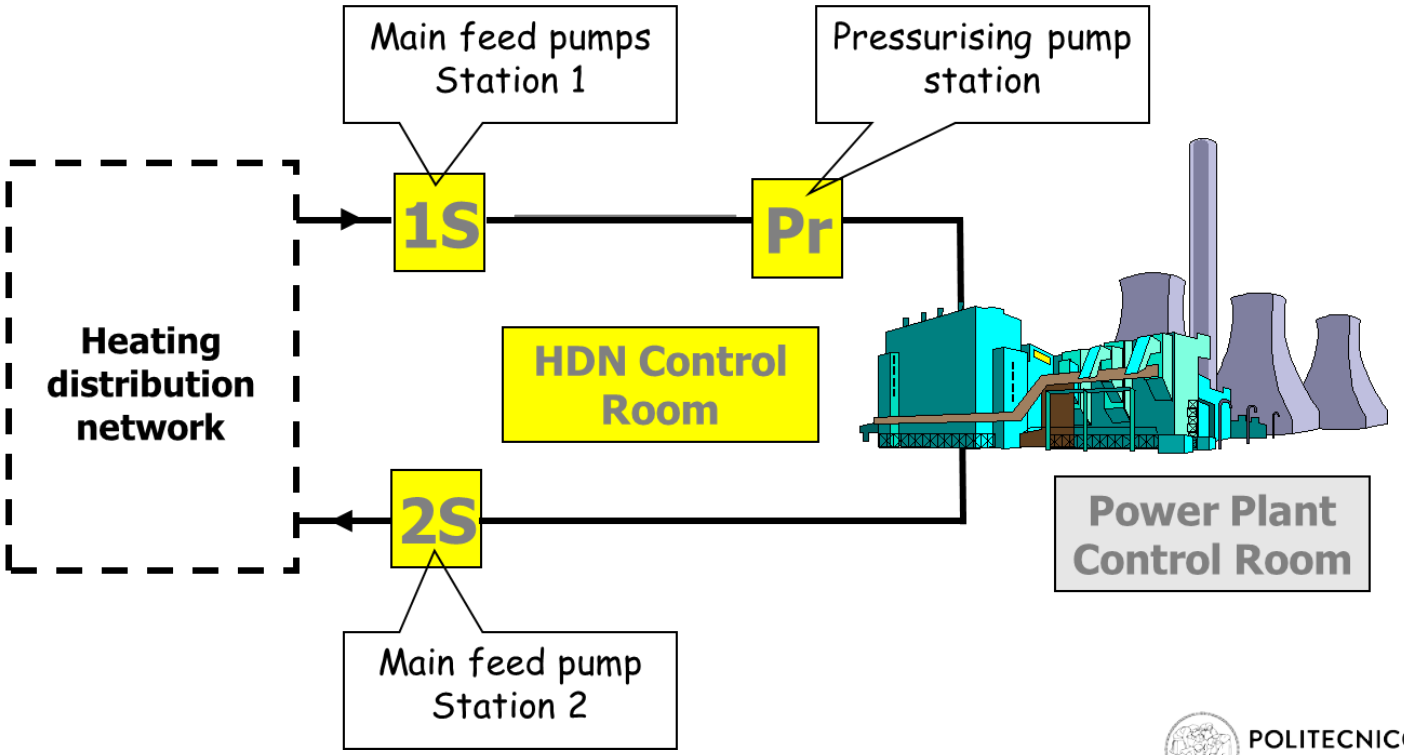
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2.0 **QUESTIONS**

Simplified scheme of the system



Main time line of the accident

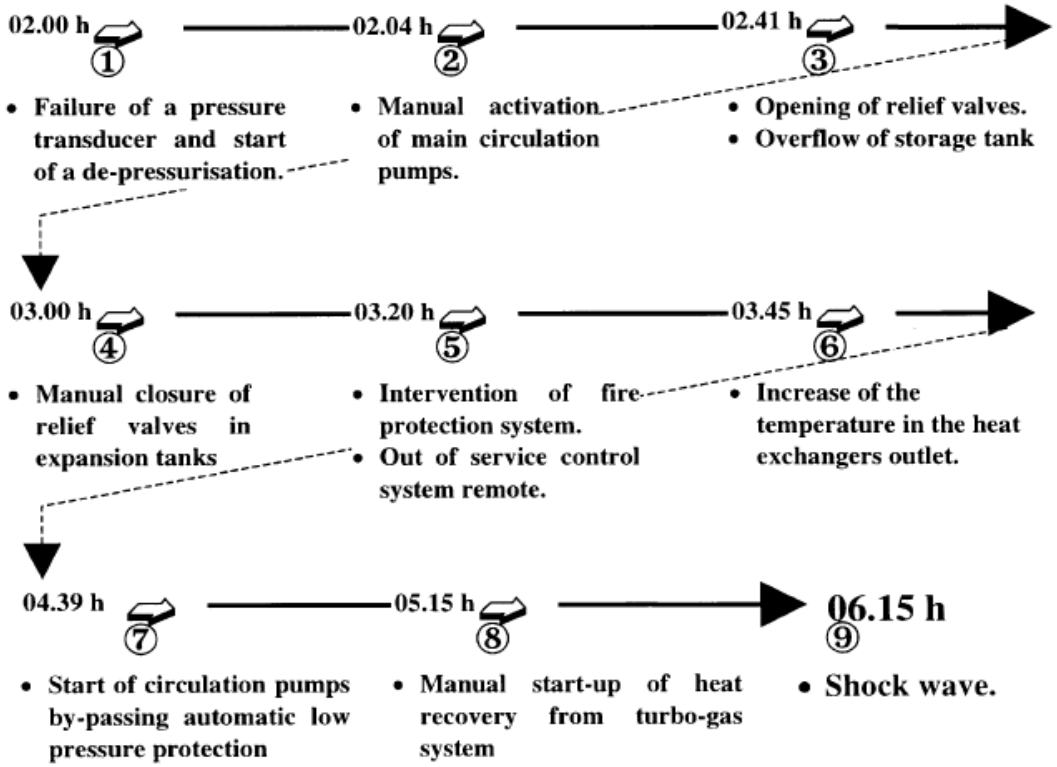
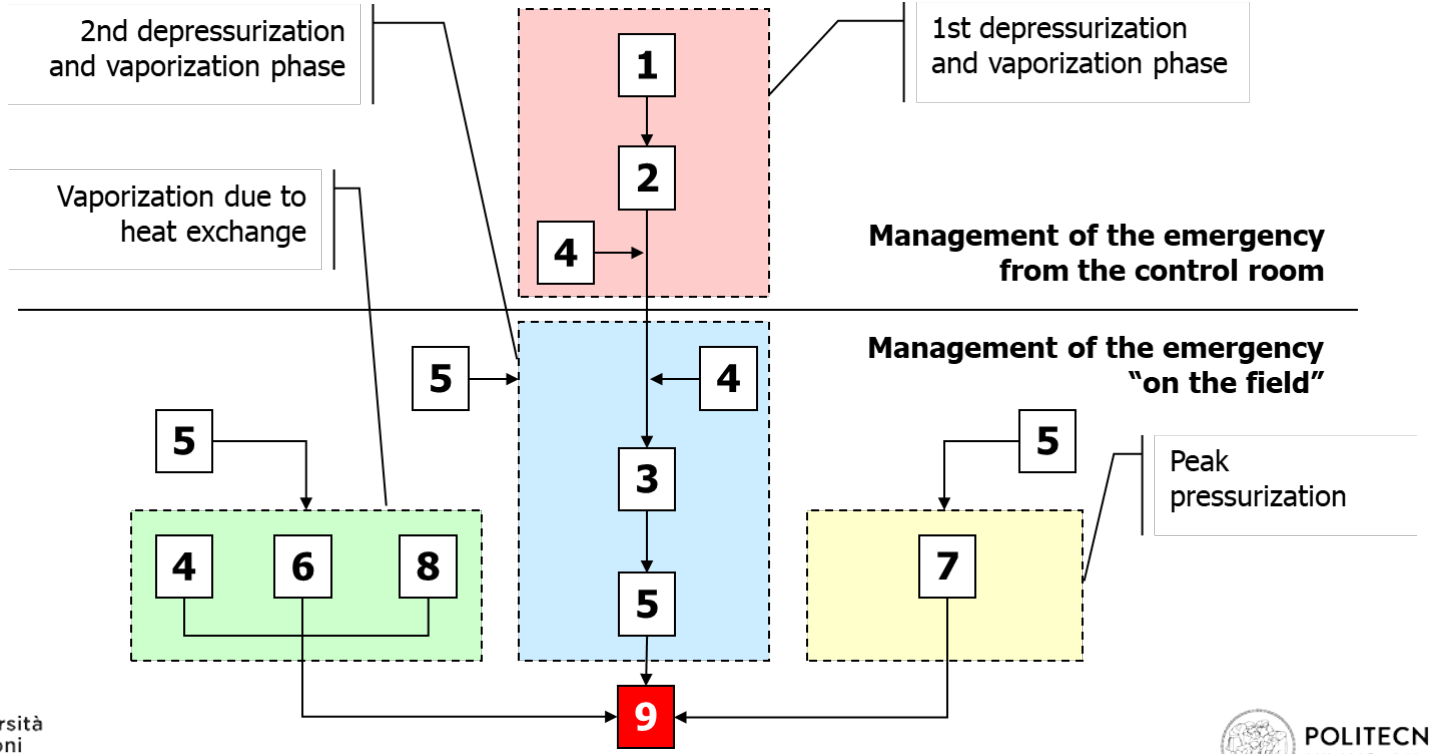


Fig. 3. Event time-line of accident.

Causal and contributing factors

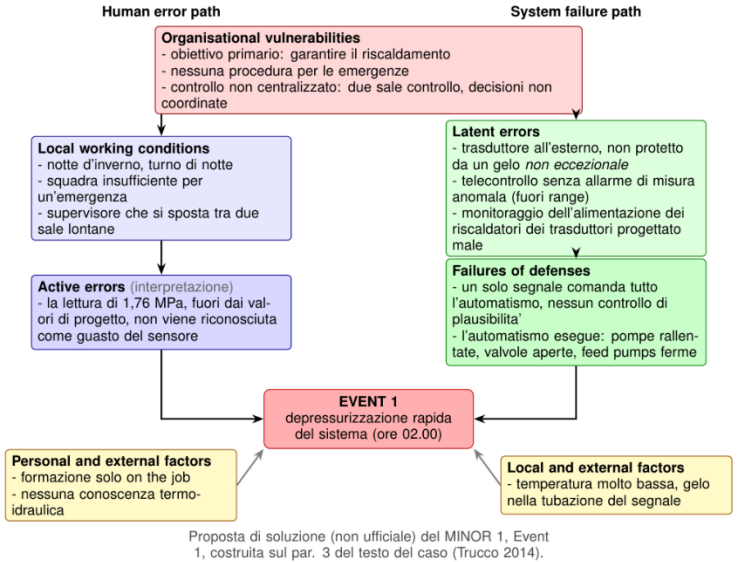


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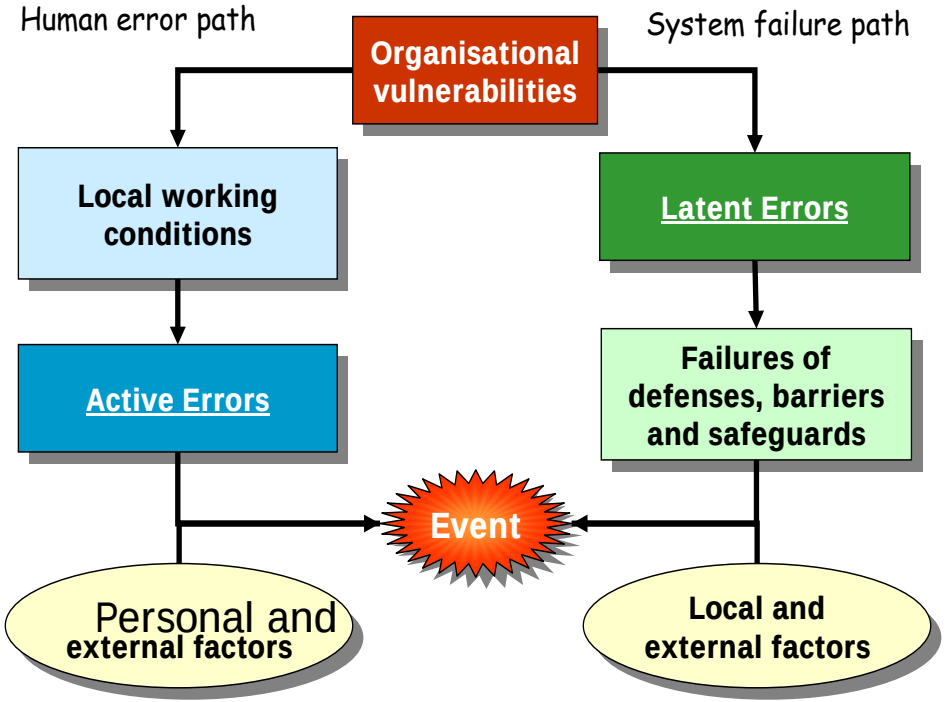
QUESTIONS

Accident Analysis

— Analyse Event 1 and 3 using the behavioural approach (Reason). Report the results by means of the following scheme



Proposta di soluzione (non ufficiale) del MINOR 1, Event 1, costruita sul par. 3 del testo del caso (Trucco 2014).



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Accident Analysis

— Event 1:

- The pressure transducer failed as consequence of a very low temperature (however, not exceptional) reached during a winter's night, which also caused freezing of the signal transmission piping. As a consequence of this failure, an erroneous pressure indication of 1.76 MPa, outside the design values, was reached, while, in reality the pressure in the system was normal. The entire automatic control system started reacting to the erroneous signal. The pressurising pump revolutions were reduced, the relief valves of the pressurising system and the valves of the expansion tanks were opened, and the main feed pumps were also stopped. A rapid transient depressurisation started.

— Event 3:

- The attainment of the maximum level in the storage tank resulted in water discharge through the relief valves. Unfortunately, the emergency system was not sufficiently dimensioned to fully discharge the water flow and some boiling water started to flood the pumping room.

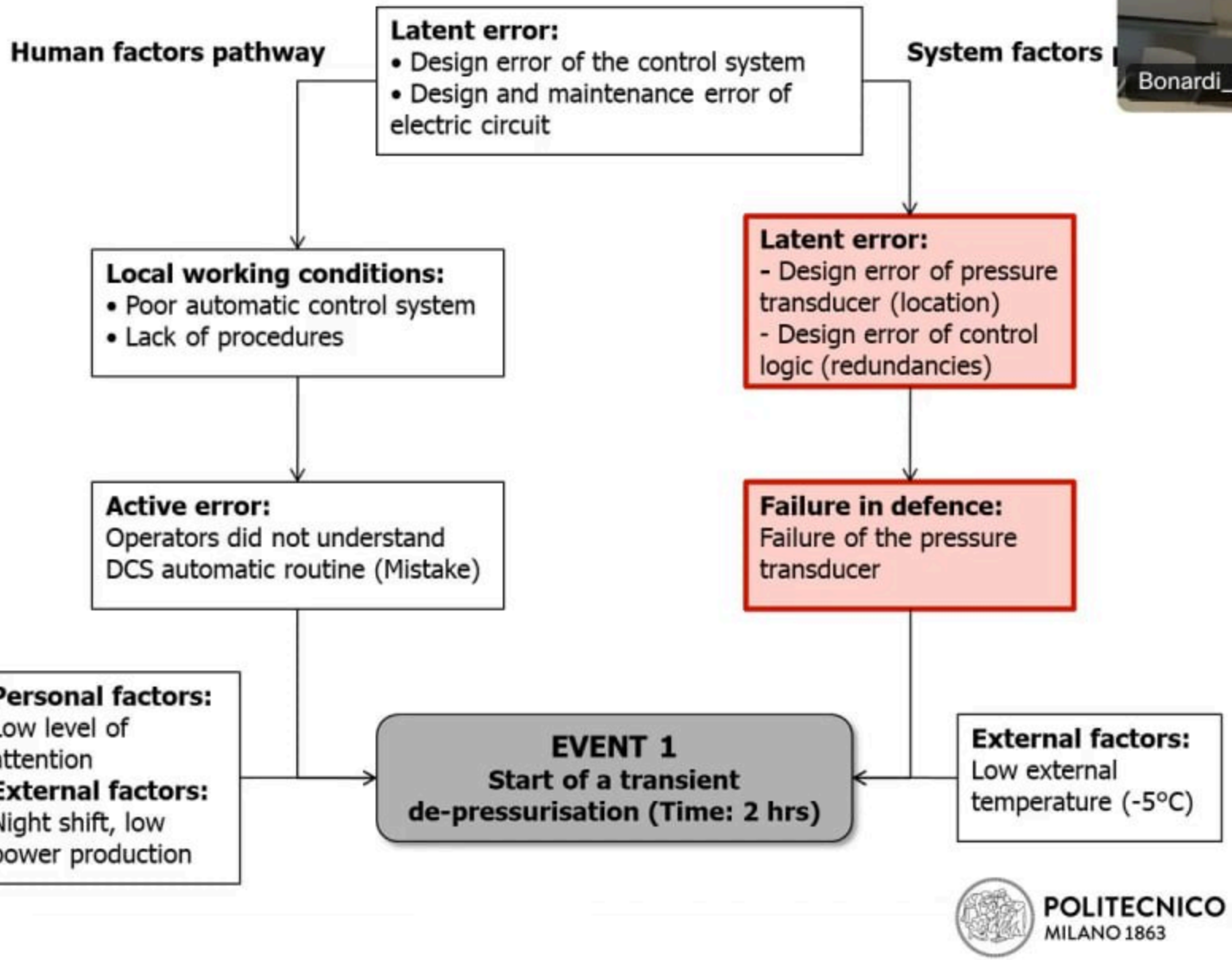


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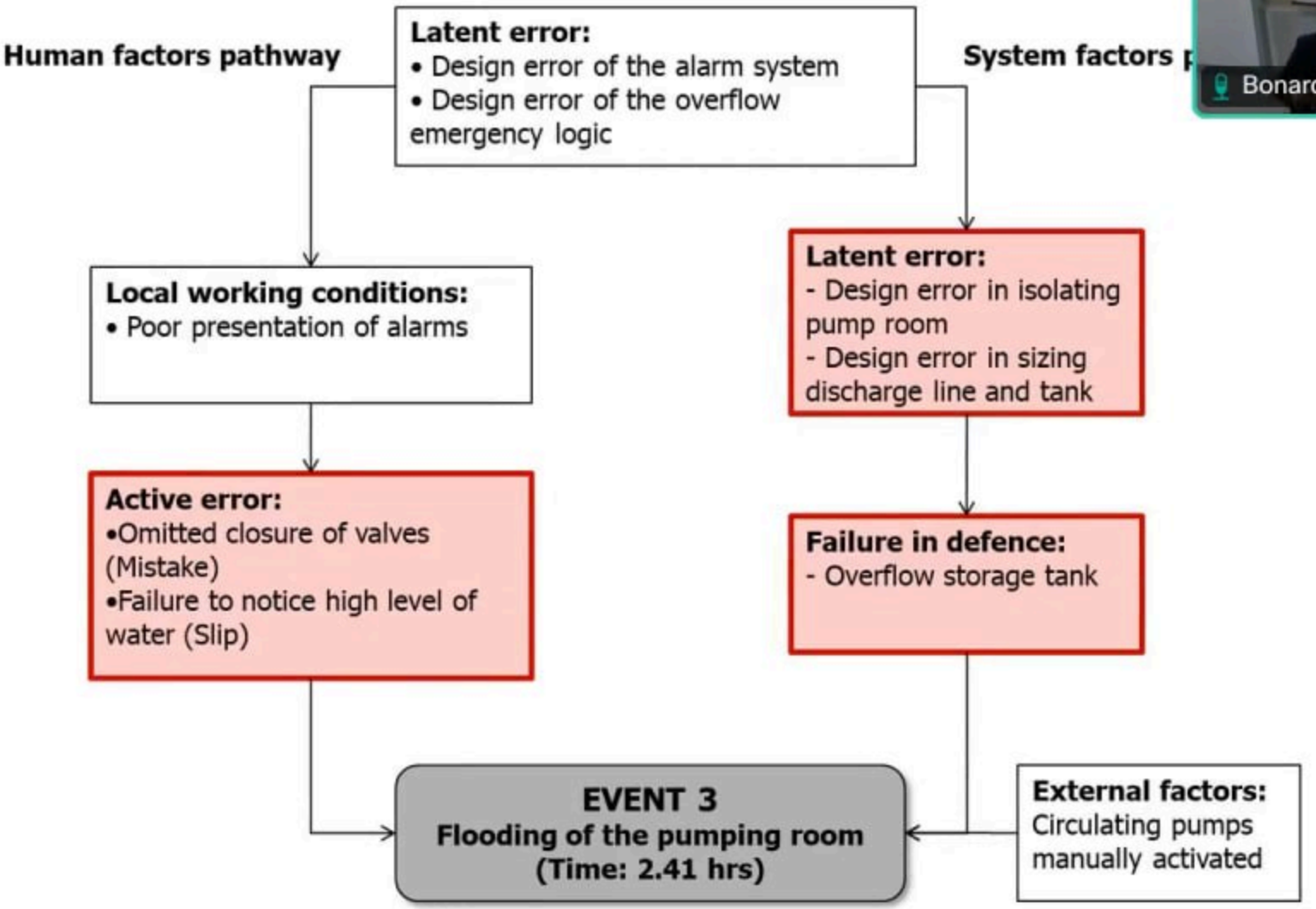
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Event 1



Bonardi_13_T.0.3

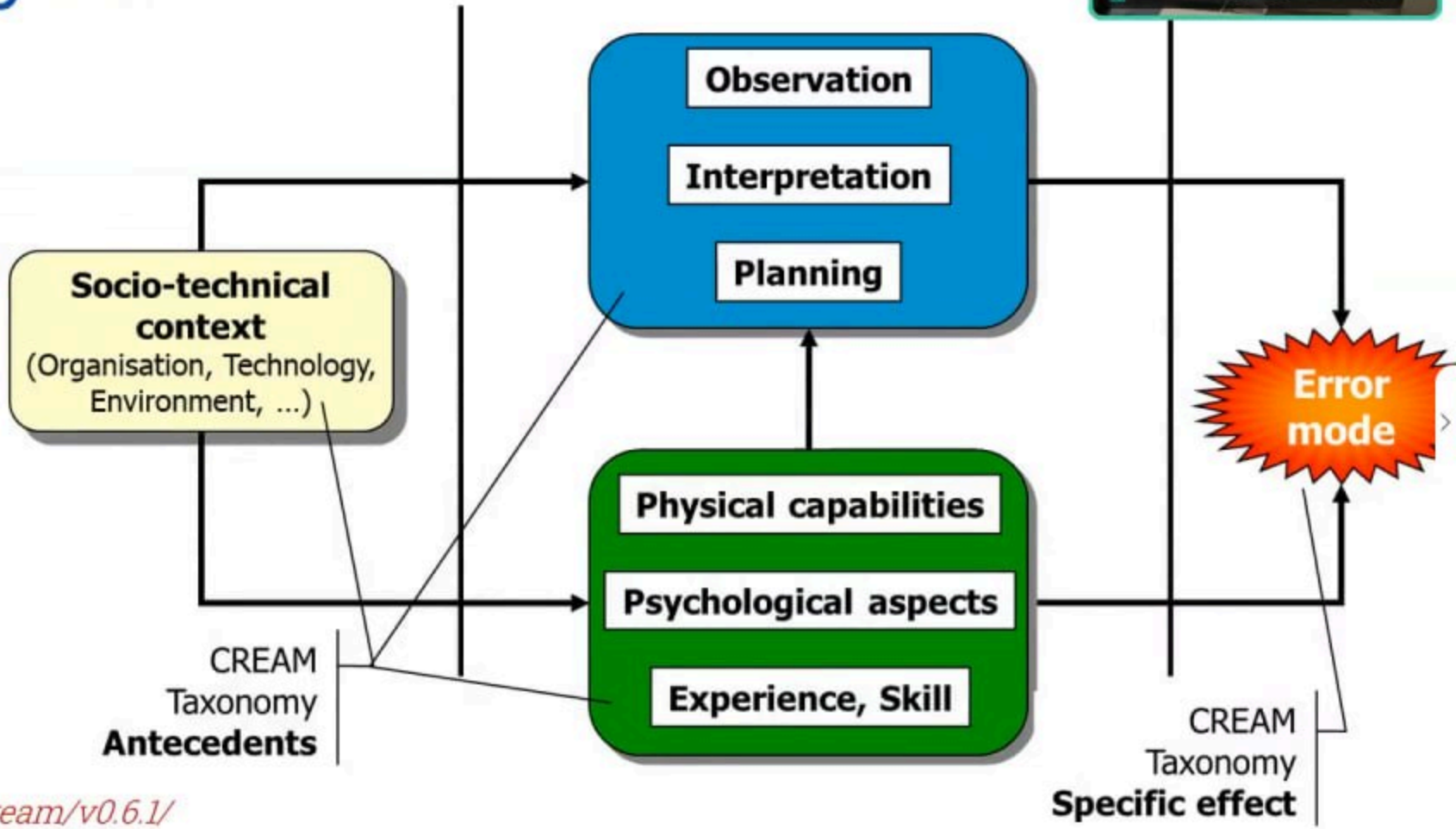
Event 3



Bonardi_13_T.0.3
48:14 / 1:15:40 1X
Bonardi_13_T.0.3

Accident analysis

— Analyse Event 1 and 3 using the behavioural approach (Reason). Report the results by means of the following scheme



http://www.ews.uiuc.edu/~serwy/cream/v0.6.1/

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Bonardi_13_T.0.3
1:05:10 / 1:15:40
1X
Bonardi_13_T.0.3